Fourier transform summary. Digital filters

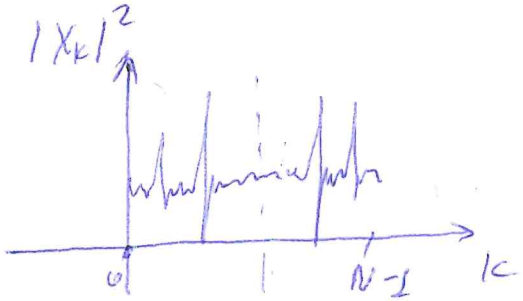
DFT : $\mathbb{C}^N \to \mathbb{C}^N$

(discrete finite) (discrete finite)

$$X_k = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x_n \exp\left(-j\frac{2\pi}{N} nk\right), \quad k = 0, \dots, N-1$$

$$x_n = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_k \exp\left(j\frac{2\pi}{N} nk\right), \quad n = 0, \dots, N-1$$

$x_n \in \mathbb{R}$



DTFT discrete infinite $\longleftrightarrow$ continuous finite
(cont. periodic infinite)

$$X(\omega) = \sum_{n=-\infty}^{+\infty} x[n] \exp(-j\omega n), \quad \omega \in \mathbb{R}$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) \exp(j\omega n), \quad n \in \mathbb{Z}$$

$$X(\omega + 2\pi k) = \sum_n x[n] \exp(-j(\omega + 2\pi k)n) =$$

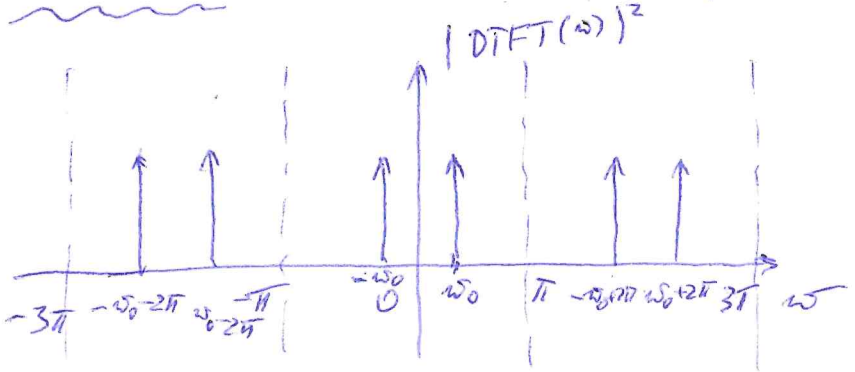
$$= \sum_n x[n] \exp(-j\omega n) \underbrace{\exp(-j2\pi k n)}_{=1} = X(\omega)$$

$\Rightarrow X(\omega)$ – periodic

Example $x(t) = \cos(\omega_0 t) \in \mathbb{R}$, $x[n] = x(nT)$, $T = \frac{1}{f_s}$,

f_s – discr. freq, Hz

$\omega = 2\pi \frac{f}{f_s}$, f – Hz

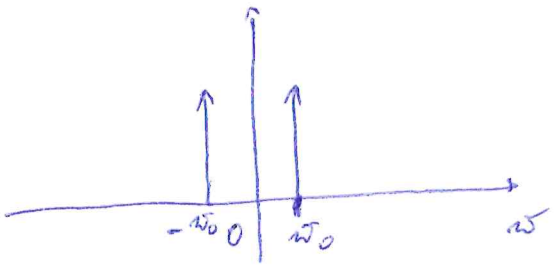


FT infinite continuous $\longleftrightarrow$ infinite cont.

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\exp(-j\omega t)\,dt, \quad \omega \in \mathbb{R}$$

$$x(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\omega)\exp(j\omega t)\,d\omega, \quad t \in \mathbb{R}$$

Example $x(t) = \cos\omega_0 t$



if $x(t) \in L^1$:

$$\int_{-\infty}^{+\infty} |x(t)|\,dt < +\infty$$

$\Rightarrow X(\omega)\ \exists$ as usual integral

$\cos(\omega_0 t),\ 1,\ \mathrm{sinc}(t) \notin L^1$

$$\int \varphi(\omega)X(\omega)\,d\omega = \int \varphi(\omega)\int x(t)\exp(-j\omega t)\,dt\,d\omega =$$

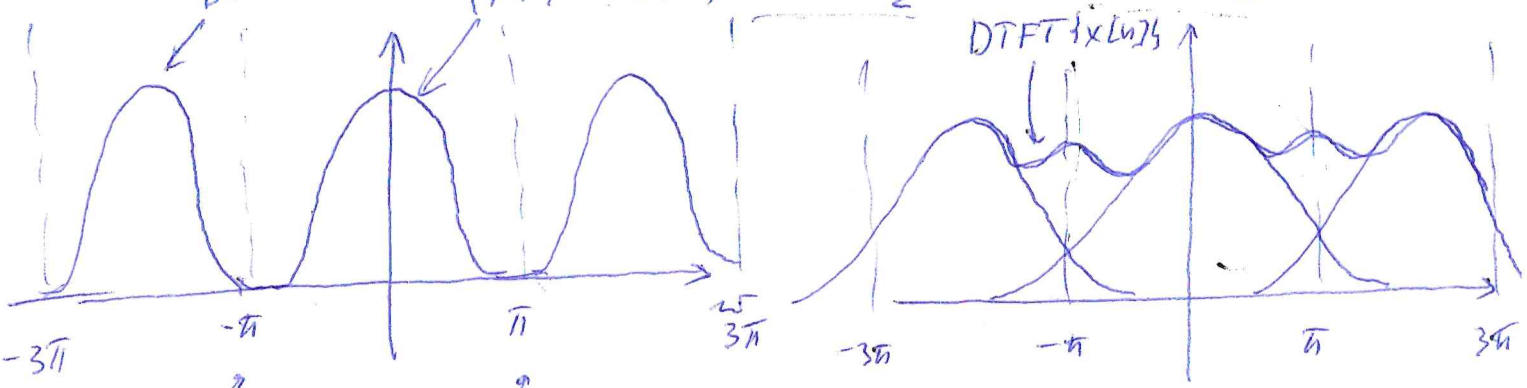
$\varphi(\omega)$ – quickly decreasing function, e.g. $\exp(-a\omega^2)$

$$= \int x(t)\underbrace{\int \varphi(\omega)\exp(-j\omega t)\,d\omega\,dt}_{<+\infty}$$

$X(\omega)$ – operator applied to $\varphi(\omega)$

Aliasing $x(t)$, $x[n] = x(nT)$, $T = \frac{1}{f_s}$, f_s – discr. freq, Hz

$\mathrm{DTFT}\{x[n]\}$ $\quad |FT\{x(t)\}|^2$, $f_{max} < \frac{f_s}{2}$ $\quad f_{max} > \frac{f_s}{2}$



$x(t) = FT^{-1}(\mathrm{DTFT}\{x(nT)\} \cdot I[-\pi, \pi])$

$x(t)$ can't be restored from $x[n]$

44.1 kHz Audio $\longrightarrow$ low pass filter $\longrightarrow$ 16 kHz Audi
(cut-off all freq. $\geq$ 8 kHz)

Spectral leakage

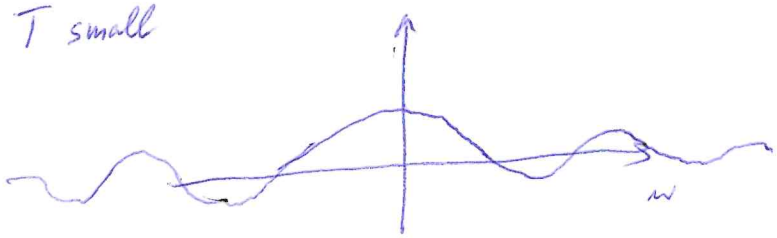
$$f(t) = \cos \omega_0 t \quad , \quad g(t) = \begin{cases} 1, & -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0, & \text{otherwise} \end{cases}$$

$$\hat{f}(t) = f(t)\, g(t)$$

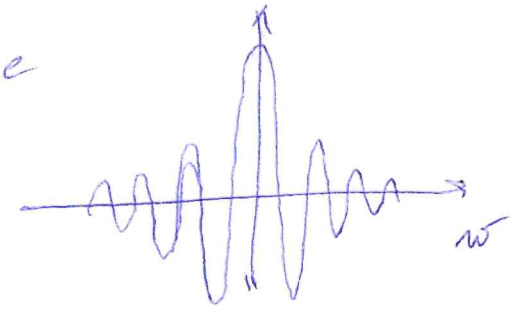
$$FT\{\hat{f}\} = FT\{f\} * FT\{g\}$$

$$FT\{g\}(\omega) = T \operatorname{sinc}\left(\frac{\omega T}{2}\right)$$

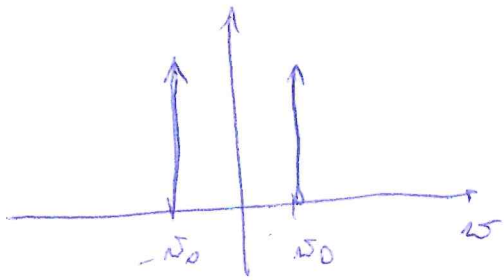
T small

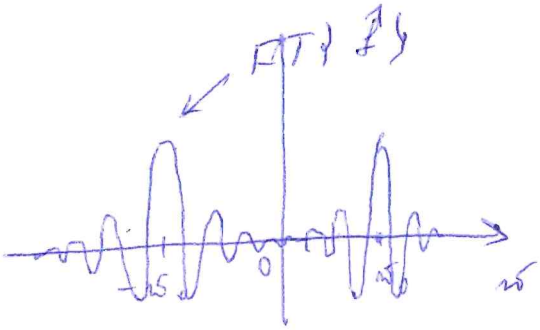


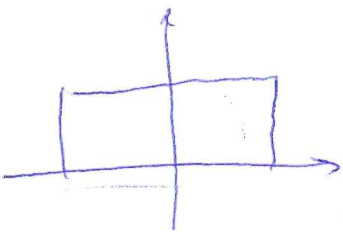
T large



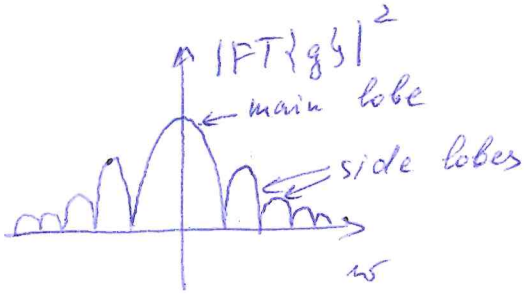
$FT\{f\}$

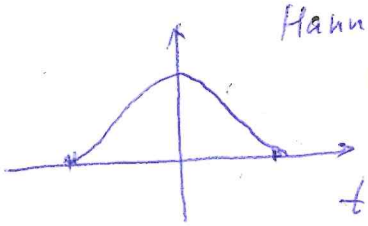






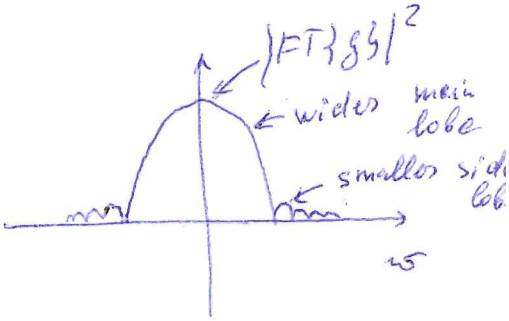
$\longrightarrow$





Hann window,
Hamming window, ...

$\longrightarrow$



Digital Filters

Filtration $y[n] = F\{x[n]\}$

F - LTI : 1) $F\{x_1[n] + x_2[n]\} = F\{x_1[n]\} + F\{x_2[n]\}$

2) $F\{x[n-k]\} = y[n-k]$

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

$h[n] = F\{\delta[n]\}$ - impulse response

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

$$y[n] = F\{x[n]\} = \sum_{k=-\infty}^{\infty} x[k] \underbrace{F\{\delta[n-k]\}}_{h[n-k]} = \sum_{k=-\infty}^{\infty} x[k]h[n-k] =$$

$$= \sum_{k=-\infty}^{+\infty} x[n-k]h[k] =$$

$$= h * x$$

$x[n] = \exp(j\omega n)$

$$y[n] = F\{x[n]\} = \sum_k h[k]x[n-k] = \sum_k h[k]\exp(j\omega(n-k))$$

$$= \exp(j\omega n) \underbrace{\sum_k h[k]\exp(-j\omega k)}_{H(\omega) \in \mathbb{C}} = H(\omega)\exp(j\omega n)$$

Types of filters

1. Finite Impulse Response (FIR) $h[n] = 0 \ \forall n > M$

$$y[n] = \sum_{k=0}^{M} h[k]x[n-k], \quad M \text{ - filter order}$$

2. Infinite Impulse Response (IIR) $h[n] \neq 0 \ \forall n$

$$y[n] = \sum_{k=0}^{M} h[k]x[n-k] - \sum_{m=1}^{N} a_m y[n-m]$$

$y[n] = F\{x[n]\} = x * h$

$Y(\omega) = H(\omega)\, X(\omega)$ $\qquad H(\omega) = \sum_n h[n] \exp(-j\omega n)$

$|H(\omega)|$ – Magnitude Response (Amplitude)

$\operatorname{Arg} H(\omega)$ – Phase Response

Algorithm for FIR coefs:

- $|H_d(\omega)| = \begin{cases} 1, & \omega_{left} \le \omega \le \omega_{right} \\ 0, & \text{otherwise} \end{cases}$

- $h_d[n] = FT^{-1}\{H_d(\omega)\}$ – sinc

- $h_d[n] = h_d[n - \frac{M}{2}]$ – shift to the right

- $h[n] = h_d[n] \cdot w[n]$

  Hann, Hamming, ...

- truncate h to M points

Adaptive Echo Cancellation using FIR

$x[n]$ – sound from speaker

$d[n]$ – signal from microphone (echo + my voice)

① $y[n] = \sum_{k=0}^{M} h[k]\, x[n-k]$

② $e[n] = d[n] - y[n]$ – error

③ $h_{new}[k] = h[k] + \mu\, e[n]\, x[n-k] \quad \forall k$

IIR stability

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] + \sum_{m=1}^{N} a_m y[n-m]$$

$$|y[n]| \xrightarrow[n\to+\infty]{} +\infty$$

ODE: $y''(t) - 3y'(t) + 2y(t) = 0$ $\quad y(0) = 2$, $y'(0) = 3$

$$y''(t) \approx \frac{y_{n+1} - 2y_n + y_{n-1}}{T}, \quad y'(t) \approx \frac{y_{n+1} - y_n}{T}$$

$$\frac{y_{n+1} - 2y_n + y_{n-1}}{T} - 3\,\frac{y_{n+1} - y_n}{T} + 2y_n = 0$$

$$y_{n+1} = f(y_n, y_{n-1})$$

$$y(t) = \exp(t) + \exp(2t) \xrightarrow[t\to+\infty]{} +\infty$$

$$X(\omega) = \int x(t)\exp(-j\omega t)\,dt$$

Laplace transform:

$$X(s) = \mathcal{L}\{x(t)\}(s) = \int_0^{+\infty} x(t)\exp(-st)\,dt, \quad s \in \mathbb{C}$$

$$s = i\omega \Rightarrow X(s) = X(i\omega) = FT\{x(t)\}(\omega)$$

$$s = \sigma + i\omega$$

$$Y(s) = \mathcal{L}\{y(t)\}(s)$$

$$\mathcal{L}\{y'(t)\}(s) = \int_0^{+\infty} y'(t)\exp(-st)\,dt =$$

$$= y(t)\exp(-st)\Big|_0^{+\infty} - \int_0^{+\infty} y(t)(-s)\exp(-st)\,dt =$$

$$= -y(0) + s\underbrace{\int_0^{+\infty} y(t)\exp(-st)\,dt}_{Y(s)}$$

$$\mathcal{L}\{y''(t)\}(s) = s^2 Y(s) - sy(0) - y'(0)$$

$$y''(t) - 3y'(t) + 2y(t) = 0 \quad | \mathcal{L}$$

$$s^2 Y(s) - s y(0) - y'(0) - 3(sY(s) - y(0)) + 2Y(s) = 0$$

$$Y(s)(s^2 - 3s + 2) = 3y(0) + y'(0) + s y(0) = 2s - 3$$

$$Y(s) = \frac{2s-3}{s^2 - 3s + 2} = \frac{2s-3}{(s-1)(s-2)} = \frac{A}{s-1} + \frac{B}{s-2} =$$

$$= \frac{1}{s-1} + \frac{1}{s-2} \Rightarrow y(t) = \exp(t) + \exp(2t)$$

$$\mathcal{L}\{\exp(at)\}(s) = \frac{1}{s-a}$$

if $Re(s) < 0 \Rightarrow$ system is stable

if $Re(s) > 0 \Rightarrow$ system is not stable

$$s = 1, 2$$

$$x(t) = \sum_{n=0}^{\infty} x(nT)\,\delta(t - nT)$$

$$\mathcal{L}\{x(t)\}(s) = \sum_{n=0}^{\infty} x(nT) \int \delta(t - nT) \exp(-st)\,dt =$$

$$= \sum_{n=0}^{\infty} x(nT) \exp(-snT) =$$

$$= \{ z = \exp(sT) \} = \sum_{n=0}^{\infty} x[n] z^{-n}$$

z-transform $\quad X(z) := \sum_{n=0}^{\infty} x[n] z^{-n}$

$s = \sigma + i\omega \qquad$ if $Re(s) < 0 \Rightarrow$ system stable

$$z = \exp(sT) = \exp((\sigma + i\omega)T) = \exp(\sigma T)\exp(i\omega T)$$

if $\underset{\sigma}{\underset{\shortparallel}{Re(s)}} < 0 \Rightarrow |z| < 1 \Rightarrow$ system is stable

$Re(s) > 0 \Rightarrow |z| > 1 \Rightarrow$ system is not stable

Example $\quad 4y[n] = 2y[n-1] - y[n-2] + 4x[n] \quad | \; Y$

$y[n-k] \rightarrow z^{-k}Y(z)$

$$4Y(z) = 2z^{-1}Y(z) - z^{-2}Y(z) + 4X(z)$$

$$Y(z)(4 - 2z^{-1} + z^{-2}) = 4X(z)$$

tranfer function $\rightarrow \dfrac{Y(z)}{X(z)} = \dfrac{4}{4-2z^{-1}+z^{-2}} = \dfrac{4z^2}{4z^2-2z+1}$

$$z_{1,2} = \frac{2 \pm \sqrt{4-16}}{8}, \quad |z_{1,2}| = \frac{1}{2} < 1$$

$\Rightarrow$ system is stable

$z = \exp(sT) \Rightarrow$

$$\Rightarrow s = \frac{1}{T}\log z \simeq \frac{2}{T}\cdot\frac{z-1}{z+1}$$

$$Y(s) \rightarrow Y(z) = Y\left(\frac{2}{T}\frac{z-1}{z+1}\right)$$